

Supplementary Material

Machine Learning-Based Predictive Autofocus for High-Resolution OpenFlexure Microscopy Using Representative Defocus Ranges

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This document accompanies the main manuscript and repository. It provides (S1) extended methodological detail, (S2) full numerical data tables underlying Figures 1–6 of the manuscript (reported only as bar charts in the main text), and (S3) supplementary notes on scope and limitations.

S1. Extended Methods

S1.1 Engineered feature set

Six features were derived from a pair of images acquired at the current stage position and at a fixed offset of 200 motor steps (10 μm) along the Z-axis:

S/NO	Feature	Definition
1	Z_prev	Current Z-position of the stage (absolute positional context)
2	variance_ratio	VoL(true) / VoL(predicted)
3	variance_sq	VoL(true) ²
4	direction	Sign of intended stage movement (+1 / -1)
5	is_improving	Binary flag: 1 if sharpness increased relative to the previous position, else 0
6	absolute_variance	VoL(true) – VoL(predicted)

Raw VoL values of the true/predicted images were used only to compute these six derived quantities and were **not** themselves passed to the models as separate inputs.

Variance of Laplacian (VoL):

$$\nabla^2 I(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

$$\text{VoL} = \text{Var}(\nabla^2 I(x, y))$$

S1.2 Hyperparameter search spaces (grid search, 5-fold CV)

SVR (RBF kernel):

Parameter	Search space
C	0.1, 1, 10, 100, 1000
γ	0.001, 0.01, 0.1, 1, 10
ϵ	0.01, 0.1, 1, 10

Random Forest:

Parameter	Search space
Number of trees (T)	10, 50, 100, 200
Max depth	None, 10, 20, 30, 50
Min samples per split	2, 5, 10, 20

Selection criterion: minimum validation-fold MSE. Features were standardized (zero mean, unit variance) for SVR only; Random Forest used raw features (tree splits are invariant to monotonic transforms).

S1.3 Dataset partitioning

Split	Samples
Total dataset	1,040 images (8 fields of view)
Training (80%)	832
Test (20%, held out)	208

Ground truth established via 4-stage hierarchical search: ultra-coarse (1000 μm range, 50 μm step) $\rightarrow$ coarse (200 μm range, 10 μm step) $\rightarrow$ fine (20 μm range, 2 μm step) $\rightarrow$ verification (± 1 μm).

S1.4 Hardware/optical configuration

Component	Specification
Compute	Raspberry Pi 4 Model B
Camera	Raspberry Pi Camera Module V2, Sony IMX219, 8 MP, 1.12 μm pixel size

Component	Specification
Acquisition resolution	640 × 480 px
Objective (blood smears)	100× oil immersion, NA 1.25 (Olympus UPlanFLN)
Objective (cancer tissue)	40× dry, NA 0.65, FoV 320 × 240 μm
Stage resolution	0.05 μm / motor step (calibrated)
Illumination	White LED, Köhler illumination, brightfield transmission

S2. Full Numerical Results (underlying Figures 1–6)

The main manuscript reports these results as grouped bar charts (Figs. 1–6). The tables below give the exact plotted values for reproducibility and reuse, grouped by sample type and initial defocus (motor steps). All values are means across $n = 7$ –10 trials per condition; error bars (SD) are shown in the corresponding figures, not reproduced here.

Table S1 — Final Variance of Laplacian (VoL, a.u.) — Fig. 1

Sample	Defocus (steps)	SVR	Random Forest	Custom AF	Inbuilt AF
Stained	500	115	98	110	108
Stained	1000	238	239	223	216
Stained	1500	70	69	109	49
Unstained	500	29.55	38.00	38.14	34.08
Unstained	1000	38.19	39.15	42.88	42.44
Unstained	1500	41.67	44.26	53.78	11.36
Cancer tissue	500	11.52	12.46	13.07	12.20
Cancer tissue	1000	10.76	15.68	26.74	9.55
Cancer tissue	1500	12.44	14.41	25.88	8.67

Table S2 — Autofocus Execution Time (s) — Fig. 2

Sample	Defocus (steps)	SVR	Random Forest	Custom AF	Inbuilt AF
Stained	500	2.66	3.13	57.19	6.37
Stained	1000	2.80	2.12	66.08	6.49
Stained	1500	4.70	4.36	88.31	6.18
Unstained	500	3.36	3.56	118.08	7.40
Unstained	1000	4.25	4.30	68.09	5.78
Unstained	1500	4.30	4.53	70.51	5.74
Cancer tissue	500	4.20	4.18	47.08	6.06
Cancer tissue	1000	2.35	2.85	29.28	7.21
Cancer tissue	1500	3.47	3.86	32.91	7.16

Table S3 — Coefficient of Variation of Final Z-Position (%) — Fig. 3

Sample	Defocus (steps)	SVR	Random Forest	Custom AF	Inbuilt AF
Stained	500	0.57	0.56	0.00	0.74
Stained	1000	0.02	0.00	0.80	0.26
Stained	1500	1.34	1.14	0.00	1.23
Unstained	500	0.00	0.41	2.81	5.70
Unstained	1000	0.09	1.34	0.53	0.37
Unstained	1500	0.09	0.02	0.72	0.31
Cancer tissue	500	0.11	0.83	1.77	9.89
Cancer tissue	1000	0.16	0.76	1.85	0.39
Cancer tissue	1500	0.62	0.34	0.64	0.32

Table S4 — Relative Error to Reference Focal Plane (%) — Fig. 4

Sample	Defocus (steps)	SVR	Random Forest	Custom AF	Inbuilt AF
Stained	500	0.64	0.75	0.00	4.27
Stained	1000	2.25	2.26	0.73	3.79
Stained	1500	20.19	21.07	0.00	18.14
Unstained	500	2.12	1.33	1.87	8.42
Unstained	1000	3.41	2.78	0.49	5.12
Unstained	1500	5.72	5.78	0.56	5.34
Cancer tissue	500	3.87	5.39	1.55	16.15
Cancer tissue	1000	5.48	4.86	1.60	12.27
Cancer tissue	1500	7.81	7.54	0.44	0.97

Table S5 — CPU Utilization (%) — Fig. 5

Sample	Defocus (steps)	SVR	Random Forest	Custom AF	Inbuilt AF
Stained	500	26.20	19.10	20.90	2.20
Stained	1000	51.90	51.60	45.90	1.90
Stained	1500	21.80	19.10	20.70	1.70
Unstained	500	20.98	28.19	27.33	1.64
Unstained	1000	22.93	22.95	27.31	2.42
Unstained	1500	23.36	23.90	20.89	2.52
Cancer tissue	500	35.63	34.29	34.56	3.79
Cancer tissue	1000	26.93	34.01	32.14	7.39
Cancer tissue	1500	31.06	36.43	29.40	8.62

Table S6 — RAM Utilization (MB) — Fig. 6

Sample	Defocus (steps)	SVR	Random Forest	Custom AF	Inbuilt AF
Stained	500	0.10	0.20	1.20	2.70
Stained	1000	0.30	0.10	1.50	1.90
Stained	1500	0.20	0.40	0.50	2.00
Unstained	500	2.71	3.89	1.42	2.61
Unstained	1000	0.13	0.07	1.17	1.89
Unstained	1500	0.07	0.13	0.47	1.08
Cancer tissue	500	3.10	4.72	1.97	2.57
Cancer tissue	1000	3.42	3.00	1.78	2.51
Cancer tissue	1500	1.89	1.55	1.90	4.73

Table S7 — Model Training and Cross-Validation Summary

Model	Test MAE (steps)	Test MAE (μm)	Test RMSE (steps)	Test RMSE (μm)	Test R^2	5-fold CV mean R^2
Random Forest	137.44	6.87	228.16	11.41	0.9814	0.9831
SVR (RBF)	224.87	11.24	291.78	14.58	0.9696	0.9656

Conversion assumes a calibrated displacement of 0.05 μm per motor step.

S3. Supplementary Notes

S3.1 Scope and reproducibility caveats

- All values in Tables S1–S6 were digitized directly from the manuscript’s own bar-chart figures (data labels shown on the plots); no values were estimated or interpolated.
- Sample sizes per condition were $n = 7\text{--}10$ trials; the tables above report means only. Refer to the manuscript’s figures for standard-deviation error bars.

- CV and relative-error values of exactly 0.00% for Custom AF reflect its role as the exhaustive-search ground-truth method in several conditions, not literal zero-variance measurement.

S3.2 Known limitations (carried from manuscript)

1. Each sample type was represented by a comparatively small number of fields of view and trials; broader staining protocols, tissue types, and cell densities remain to be tested.
2. The accuracy degradation observed at 1500 motor steps (stained sample) indicates the effective operating range of the trained regression models is bounded below this offset; extending it will likely require additional training data at larger defocus magnitudes or a hybrid predictive + brief confirmatory-search strategy.
3. Resource utilization (CPU/RAM) was measured on a single embedded hardware configuration (Raspberry Pi 4); results may differ on other single-board computers.
4. Generalization findings on unstained and tissue-derived specimens would benefit from replication across a larger, more diverse specimen panel.

S3.3 Data and code availability

The image dataset (1,040 labelled images, 8 fields of view) and trained model weights are available from the corresponding author upon reasonable request. Code for feature extraction, model training, and the live autofocus runtime is provided in this repository (see main README.md).

Citation

If you use this supplementary data, please cite the main manuscript:

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